Type MS · Pixels, resolution and colour depth

Answer 1 of 4 · Question 3 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part b.ii in this mark scheme extract

3(a)	Award up to **two** marks for a linked explanation, such as: • If the data is incomplete/inaccurate (1) the answers from the model might not be right (1) • If the assumptions the model is based on are inaccurate (1) the answers from the model might be incomplete/inaccurate (1)	Award one mark for identifying a problem and one mark for a reason There must be a clear cause and consequence/effect	2

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for any of: • Pixels per inch/ppi (1) • Pixels per centimetre/ppc (1) • Dots per inch/dpi (1)	Ignore units cm^2 $inches^2$	1

Type MS · Pixels, resolution and colour depth

Answer 1 of 4 · Question 3 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	Award **two** marks for a linked description such as: • Each pixel/colour in the image (1) will require more bits to store (1)	Giving for formula file_size = width*height*colour_depth on its own is not enough	2

Question Number	Answer	Additional Guidance	Mark
3(c)	Award **two** marks for a linked description such as: • System software is used to maintain/run the operation of the system/hardware (1) whereas application software allows the user to carry out a task (1)	For both marks both categories must be addressed Allow clear examples	2

Type MS · Pixels, resolution and colour depth

Answer 1 of 4 · Question 3 part b.ii · 2 marks

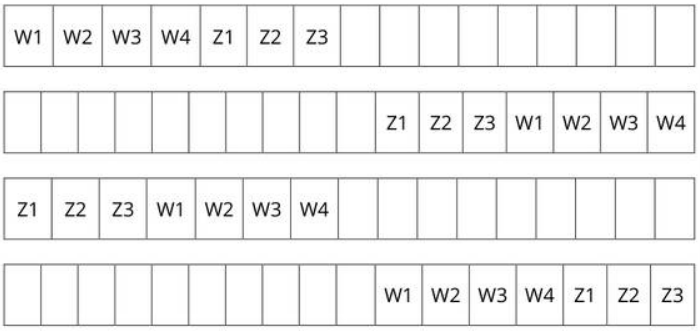
2023 JUN · Paper 01 · extract 3 of 5 · find part b.ii in this mark scheme extract

<table>
<tr><th>Question number</th><th>Answer</th><th>Additional guidance</th><th>Mark</th></tr>
<tr><td>3(d)</td><td>Award one mark for each of:

• All file blocks of each file are in numeric sequence/one after another in numeric order (1)

• All blank space is together at the front/at the end (1)

Examples:

 
</td><td>Note: Order of letter and number does not matter. W1 or 1W are equally acceptable</td><td>2</td></tr>
</table>

Type MS · Pixels, resolution and colour depth

Answer 1 of 4 · Question 3 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)	Award up to **two** marks for a linked explanation such as: • The file size will not be decreased/will increase (1) because there are very few/no repeating patterns / because a run length of one would be added to each digit (1)		2

Question Number	Answer	Additional Guidance	Mark
3(f)	Award **one** mark for each correct cell:		4

Task	Translator
A guessing game that can be used on different computing platforms	Interpreter (1)
A screen driver for a new smartphone	Assembler (1)
A new version of a spreadsheet program for sale next year	Compiler (1)
Control software for an embedded system inside a new washing machine	Assembler (1)

Type MS · Pixels, resolution and colour depth

Answer 1 of 4 · Question 3 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 5 of 5 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Pixels, resolution and colour depth

Answer 2 of 4 · Question 1 part g.i · 1 mark

2021 JUN · Paper 01 · extract 1 of 2 · find part g.i in this mark scheme extract

Number			
1(a)	**The only correct answer is B** ***A*** *is not correct because computers only understand binary* ***C*** *is not correct because Hex values will be held as binary so take up the same space* ***D*** *is not correct because Hex values are held as binary and so take the same time to execute*		1

Question Number	Answer	Additional Guidance	Mark
1(b)	Award **one** mark for each of: • 5 • A	Must be in the correct order	2

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is B** ***A*** *is not correct because this is the number of binary patterns that can be represented by 7 bits* ***C*** *is not correct because this is the number of binary patterns that can be represented by 9 bits* ***D*** *is not correct because this is the number of binary patterns that can be represented by 10 bits*		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each nibble 0110 0101		2

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark from: • 01000 (1) • 1000 (1)		1

Type MS · Pixels, resolution and colour depth

Answer 2 of 4 · Question 1 part g.i · 1 mark

2021 JUN · Paper 01 · extract 2 of 2 · find part g.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(f)	7		1

Question Number	Answer	Additional Guidance	Mark
1(g)(i)	Award **one** mark from: • The smallest block (of colour) in an image (1) • A single dot in a picture (1) • A picture element (1) • A single element of a digital screen (1)		1
1(g)(ii)	Award **one** mark from: • 8 x 9 (1) • 9 x 8 (1)	Accept 72	1
1(g)(iii)	4		1

Question Number	Answer	Additional Guidance	Mark

Type MS · Pixels, resolution and colour depth

Answer 3 of 4 · Question 3 part b.i · 1 mark

2023 JUN · Paper 01 · extract 1 of 5 · find part b.i in this mark scheme extract

3(a)	Award up to **two** marks for a linked explanation, such as: • If the data is incomplete/inaccurate (1) the answers from the model might not be right (1) • If the assumptions the model is based on are inaccurate (1) the answers from the model might be incomplete/inaccurate (1)	Award one mark for identifying a problem and one mark for a reason There must be a clear cause and consequence/effect	2

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for any of: • Pixels per inch/ppi (1) • Pixels per centimetre/ppc (1) • Dots per inch/dpi (1)	Ignore units cm^2 $inches^2$	1

Type MS · Pixels, resolution and colour depth

Answer 3 of 4 · Question 3 part b.i · 1 mark

2023 JUN · Paper 01 · extract 2 of 5 · find part b.i in this mark scheme extract

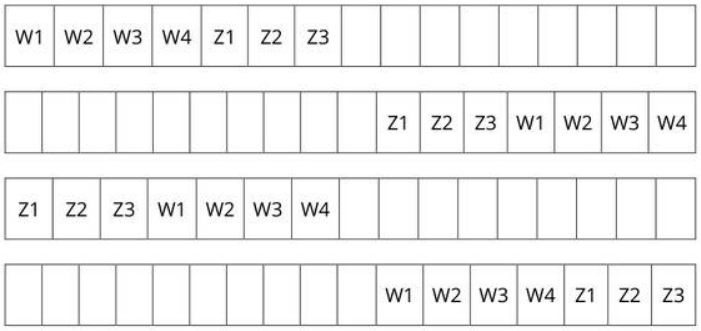
Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	Award **two** marks for a linked description such as: • Each pixel/colour in the image (1) will require more bits to store (1)	Giving for formula file_size = width*height*colour_depth on its own is not enough	2

Question Number	Answer	Additional Guidance	Mark
3(c)	Award **two** marks for a linked description such as: • System software is used to maintain/run the operation of the system/hardware (1) whereas application software allows the user to carry out a task (1)	For both marks both categories must be addressed Allow clear examples	2

Type MS · Pixels, resolution and colour depth

Answer 3 of 4 · Question 3 part b.i · 1 mark

2023 JUN · Paper 01 · extract 3 of 5 · find part b.i in this mark scheme extract

Question number	Answer	Additional guidance	Mark
3(d)	Award **one** mark for each of: • All file blocks of each file are in numeric sequence/one after another in numeric order (1) • All blank space is together at the front/at the end (1) Examples: 	Note: Order of letter and number does not matter. W1 or 1W are equally acceptable	2

Type MS · Pixels, resolution and colour depth

Answer 3 of 4 · Question 3 part b.i · 1 mark

2023 JUN · Paper 01 · extract 4 of 5 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)	Award up to **two** marks for a linked explanation such as: • The file size will not be decreased/will increase (1) because there are very few/no repeating patterns / because a run length of one would be added to each digit (1)		2

Question Number	Answer	Additional Guidance	Mark
3(f)	Award **one** mark for each correct cell:		4

Task	Translator
A guessing game that can be used on different computing platforms	Interpreter (1)
A screen driver for a new smartphone	Assembler (1)
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Control software for an embedded system inside a new washing machine	Assembler (1)

Type MS · Pixels, resolution and colour depth

Answer 3 of 4 · Question 3 part b.i · 1 mark

2023 JUN · Paper 01 · extract 5 of 5 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Pixels, resolution and colour depth

Answer 4 of 4 · Question 3 part b.i · 1 mark

2025 JUN · Paper 01 · extract 1 of 3 · find part b.i in this mark scheme extract

3(a)(i)	Award **one** mark for each correct hexadecimal character in the correct order: • B9		2
Question	Answer	Additional Guidance	Mark
3(a)(ii)	**The only correct answer is B** *A is not correct because both are processed at the same speed by a computer* *C is not correct because both use the same amount of storage* *D is not correct because hexadecimal uses number base 16*		1
Question	Answer	Additional Guidance	Mark
3(a)(iii)	Award **two** marks for a linked description such as: • (logical/arithmetic) right shift (1) by 2 places (1)		2
Question	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for: Pixel/picture element (1)	Accept additional words. Accept incorrect spelling if word is recognisable as pixel	1

Type MS · Pixels, resolution and colour depth

Answer 4 of 4 · Question 3 part b.i · 1 mark

2025 JUN · Paper 01 · extract 2 of 3 · find part b.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark
3(b)(ii)	Award **one** mark for each of: • 16x16 pixels (1) • Multiplying (16x16) by 2 bits per pixel in the numerator for colour depth (1) • Dividing by 8 in the denominator to convert to bytes (1) • Dividing 1000 by the number of bytes used for one image (1) Example: $1000 / \frac{16 \times 16 \times 2}{8}$ bytes Allow follow-through	4 marks for any of: 1000 X 8 / 16 x 16 x 2 8000 / 256 x 2 1000 / 16 x 2 x 2 1000 / 16 x 4 8000 / 512	4
Question	Answer	Additional Guidance	Mark
3(b)(iii)	Award up to **two** marks for a linked explanation such as: • high compression ratio (1) repeated/continuous runs (of the same pixel colour) (1) • requires less storage space (1) encoding 16W for line 1 (1)	Allow valid RLE examples from the image. Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
Question	Answer	Additional Guidance	Mark
3(c)	Award **one** mark for each correct cell: Encrypted text / Plain text  RED  COT	Ignore capitalisation Must be correct spelling	2
		Total for question 3	**14**

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Type MS · Pixels, resolution and colour depth

Answer 4 of 4 · Question 3 part b.i · 1 mark

2025 JUN · Paper 01 · extract 3 of 3 · find part b.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark